Small Is Clean: Environmental Regulation and Plant Contraction in India

Megha Patnaik
LUISS Guido Carli and CEPR

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Motivation

What margins do firms use to comply with environmental regulation?

Empirical evidence from advanced economies:

- *Innovation and productivity:* Aghion, Dechezleprêtre, Hemous, Martin, and Van Reenen (2016); Calel and Dechezleprêtre (2016); Greenstone, List, and Syverson (2012); Klenow, Pastén, and Ruane (2025).
- *Relocation:* Martin, Muuls, de Preux, and Wagner (2014); Hanna (2010).
- *Employment:* Walker (2013); Fuster, Kabas, Pezone, and Roszbach (2025).

In developing countries, firms face greater frictions on margins of adjustment, such as access to capital and technology.

This Paper

- I analyse the response of firms to environmental regulation in the context of India.
- India is the world's third-largest CO_2 emitter, and its industrial emissions continue to rise while those of advanced economies fall.
- I use the *Perform, Achieve and Trade (PAT)* scheme: a plant-specific energy-intensity standard launched in 2012, covering ∼60% of industrial energy use and the template for India's 2025 carbon market (CCTS).
- I find that energy-intensive plants comply by contracting, not innovating:
  - *Contraction:* fuel 9–18%, capital 7%, employment 5–8%; largest in older plants.
  - *No innovation:* R&D, investment, and product upgrading unchanged, unlike the advanced-economy benchmark.
  - *Misallocation:* a heterogeneous wedge raises TFPR without TFPQ for fuel-intensive plants, widening revenue-productivity dispersion.

India's PAT Scheme

Policy Timeline.

- *2001:* Energy Conservation Act establishes the Bureau of Energy Efficiency.
- *2012:* PAT launched under the National Action Plan on Climate Change, as its flagship industrial market mechanism.
- *2025:* Carbon Credit Trading Scheme (CCTS) launches, India's first economy-wide carbon market.

PAT scheme structure:

- Large plants (*Designated Consumers*) in eight energy-intensive sectors (power, steel, cement, aluminium, fertilizer, paper, textiles, chlor-alkali): about 60% of industrial energy use.
- Each plant receives a plant-specific energy-intensity (SEC) reduction target, set in three-year cycles (Cycle I: 2012–15).
- Plants that beat their target earn tradable Energy Saving Certificates (ESCerts); those that miss must buy them.

CCTS (2025) inherits this architecture. The compliance unit shifts from energy to emissions intensity, but the plant-level, intensity-target design carries over.

Example: Gazette Notification

60 THE GAZETTE OF INDIA : EXTRAORDINARY [PART II—SEC. 3(ii)]

(1)	(2)	(3)		(4)
	Name, Address and State	Specific energy consumption (TOE/Ton of product)	Product Output (Ton)	Specific energy consumption (TOE/Ton of product)
7	National Aluminium Company Limited S & P Complex, At/Po: Nalconagar, Dist-Angul, Pin-759145 Odisha	5.474	383988	5.199
8	Vedanta Aluminium Limited Village - Burkhamunda, P.O. Sripura, Jharsuguda, Pin-768202 Odisha	6.405	215641	6.028
9	Hindalco Industries Limited P. O. Renukoot, Dist- Sonbhadra, Pin-231217 Uttar Pradesh	5.858	388869	5.512
10	Hindalco Industries Limited Plot No. 2, MIDC Taloja, A.V, Navi Mumbai, Dist Raigad, Pin-410208 Maharashtra	0.183	53320	0.172

(ii) Chlor-Alkali

1.	Aditya Birla Chemicals India Limited. Garhwa Road,PO: Rehla, Dist. Palamau, 822124	0.997	90,777	0.928

Each plant's target is a **percentage cut in energy intensity**:

$$r_i = \frac{\text{baseline SEC} - \text{target SEC}}{\text{baseline SEC}}.$$

National Aluminium: $5.474 \to 5.199$, a **5.0%** cut. Hindalco (Taloja): $0.183 \to 0.172$, a **6.0%** cut.

Data: Digitized Gazette

I hand-build the first plant-level dataset of PAT regulatory assignments from the set of gazette notifications.

All PAT notifications, Cycles I–VIII (2012–2023), many scanned and bilingual, plus the regulator's machine-readable target spreadsheets. I extract the English tables and validate against published plant counts.

Plant	**Sector**	**Baseline SEC**	**Target SEC**	**Cut**
National Aluminium	Aluminium	5.474	5.199	5.0%
Vedanta Aluminium	Aluminium	6.405	6.028	5.9%
Hindalco (Taloja)	Aluminium	0.183	0.172	6.0%
Grasim Industries	Chlor-alkali	0.310	0.291	6.1%
Aditya Birla Chem.	Chlor-alkali	0.997	0.928	6.9%

2,012 plant-cycle assignments: 1,553 in manufacturing (eight sectors), from ∼800 unique plants each reassigned a fresh target every cycle. Targets (SEC, TOE/ton) span 0.1–29.2%.

▸ Coverage & targets

Data: The ASI Factory Panel

I build the longest available factory-level panel from the Annual Survey of Industries (ASI, India's MoSPI): 26 years, 1998–2023, the first to extend through 2023 and over a decade of post-PAT data.

Sample. The ASI is a census of registered factories. I use the *census stratum* ($\geq$100 workers, surveyed every year with weight one): 533,419 factory-year observations. PAT plants fall overwhelmingly here, and these factories are observed every operating year, avoiding rotation attrition.

Variables. Item-level fuel expenditure (coal, petroleum, electricity, gas), output and sales, capital, employment (direct and contract), wages, intermediates, and product-level quantities and unit values; R&D from 2015.

Productivity. I compute TFPR and TFPQ following Klenow et al. (2025): gross output over total cost with energy as a separate input, keeping the energy wedge visible (vs. value-added approaches that net energy out).

Measuring Exposure to PAT

PAT targets the most fuel-intensive plants within a sector. For factory i in sector s, I use *pre-period* (2008–2011) fuel intensity, standardized within sector:

$$\mathrm{FI}_i^* = \frac{\mathrm{FI}_{i,\mathrm{pre}} - \bar{\mathrm{FI}}_s}{\sigma_s}, \qquad \mathrm{FI}_{i,\mathrm{pre}} = \frac{1}{T_{\mathrm{pre}}} \sum_{t=2008}^{2011} \frac{\mathrm{fuel}_{it}}{\mathrm{output}_{it}}.$$

- *Predetermined:* built only from pre-2012 data, so PAT cannot have moved it.
- Higher FI_i^* within a PAT sector $\Rightarrow$ more likely a Designated Consumer facing a stricter target.
- *Continuous* treatment using all plants, no binary treated/untreated split.

Empirical Strategy

I estimate a triple difference: high- vs. low-FI plants within PAT sectors, post vs. pre.

$$y_{ist} = \beta \left(\text{PAT}_s \times \text{FI}_i^* \times \text{Post}_t\right) + \alpha_i + \gamma_{st} + \varepsilon_{ist}$$

- FI_i^* is standardized pre-period fuel intensity; $\text{Post}_t = \mathbb{1}(t \geq 2012)$.
- α_i are factory fixed effects; γ_{st} sector-by-year fixed effects; SE clustered by factory.
- γ_{st} absorbs all sector-wide shocks (energy prices, trade, demand); α_i absorbs fixed plant traits.
- Single treatment date (2012): no staggered-TWFE weighting problems.

Results: Effect of PAT on Fuel Use, Inputs, and Output

	(1) Fuels	(2) Fuel Int	(3) K	(4) Emp	(5) W	(6) Y	(7) Sales
PAT $\times$ FI_{pre} $\times$ Post	-0.116***	-0.094***	-0.072***	-0.076***	-0.091***	-0.022	-0.037*
	(0.026)	(0.018)	(0.023)	(0.028)	(0.025)	(0.019)	(0.022)
Factory FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sector $\times$ Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	254,226	254,226	254,399	254,385	254,399	254,399	254,399

Notes: All dependent variables are in logs. Each column reports a separate triple-difference regression of the column dependent variable on PAT sector $\times$ standardized pre-period fuel intensity $\times$ Post from equation (1). K = capital, Emp = employees, W = total wages, Y = output, Sales = gross sales; Y and Sales are nominal gross output measured as the ex-factory value of production in rupees, with sector$\times$year fixed effects absorbing common output-price movements. The capital-labor ratio and wage per worker (untabulated) are not significantly affected. Sample includes all census-stratum factories (PAT and non-PAT). Factory fixed effects absorb PAT sector status and baseline fuel intensity; sector$\times$year fixed effects absorb sector-level shocks. Standard errors clustered at the factory level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

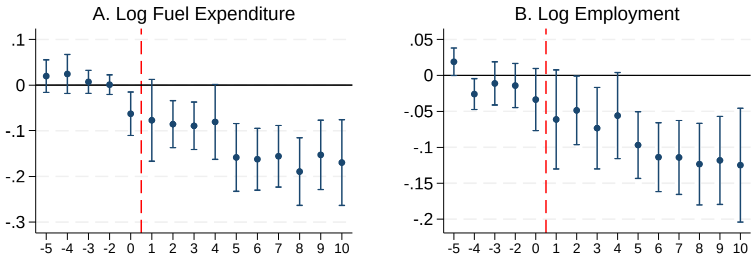
Results: Dose-Response by Pre-Period Fuel Intensity Quartile

	(1) Log Fuels	(2) Log Emp	(3) Log Output	(4) Log Capital
Q2 × Post	−0.0304	0.0345	0.0350	−0.0158
	(0.0345)	(0.0272)	(0.0327)	(0.0323)
Q3 × Post	−0.1174***	−0.0559**	−0.0555*	−0.0946***
	(0.0335)	(0.0265)	(0.0315)	(0.0307)
Q4 × Post	−0.2257***	−0.1462***	−0.1070***	−0.1553***
	(0.0401)	(0.0281)	(0.0359)	(0.0341)
Factory FE	Yes	Yes	Yes	Yes
Sector × Year FE	Yes	Yes	Yes	Yes
Observations	502,673	503,301	503,315	503,315

Notes: Plants in PAT sectors are split into quartiles of pre-period fuel intensity within each sector. Q1 (lowest intensity) is the omitted category. Each coefficient is the interaction of the quartile indicator with Post (year $\geq$ 2012). The sample includes all census-stratum factories (PAT and non-PAT sectors); non-PAT plants and PAT plants without valid pre-period fuel intensity are absorbed into the Q1 reference category. The sample size is larger than the pooled triple-difference tables because the dummy specification does not require valid pre-period fuel intensity for every plant. Factory fixed effects absorb PAT sector status and baseline fuel intensity; sector×year fixed effects absorb sector-level shocks. Standard errors clustered at the factory level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Event Study: Fuel and Employment

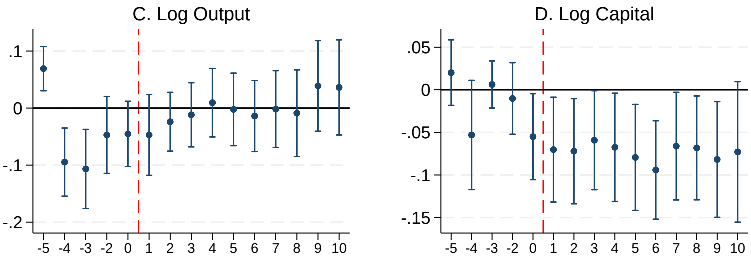
$$y_{ist} = \sum_{k=-5}^{+10} \beta_k \left(\text{PAT}_s \times \text{FI}_i^* \times \mathbb{1}(t = 2012 + k)\right) + \alpha_i + \gamma_{st} + \varepsilon_{ist}$$



Notes: Each panel plots coefficients β_k from the event-study specification with factory and sector-by-year fixed effects. The treatment is PAT sector $\times$ standardized pre-period fuel intensity $\times$ year dummies. Base year is 2011 ($k = -1$). The sample includes all census-stratum factories (PAT and non-PAT sectors). Standard errors clustered at the factory level; vertical bars show 95% confidence intervals. Dashed red line marks 2012 (PAT Cycle I implementation).

Event Study: Output and Capital

$$y_{ist} = \sum_{k=-5}^{+10} \beta_k \left(\text{PAT}_s \times \text{FI}_i^* \times \mathbb{1}(t = 2012 + k) \right) + \alpha_i + \gamma_{st} + \varepsilon_{ist}$$



Years relative to PAT (2012). Dashed red line marks implementation.
95% CIs from factory-clustered SEs. Base year: 2011 (k=-1).

Notes: Each panel plots coefficients β_k from the event-study specification with factory and sector-by-year fixed effects. The treatment is PAT sector $\times$ standardized pre-period fuel intensity $\times$ year dummies. Base year is 2011 ($k = -1$). The sample includes all census-stratum factories (PAT and non-PAT sectors). Standard errors clustered at the factory level; vertical bars show 95% confidence intervals. Dashed red line marks 2012 (PAT Cycle I implementation).

Mechanism: Contraction, Not Innovation

The contraction comes with *no* technology-adoption or upgrading response:

- **R&D:** -0.019 (n.s.): no innovation response.
- **Gross investment:** $+0.028$ (n.s.): no capital upgrading.
- **Product scope and unit values:** unchanged; no shift in product mix or quality.
- **Exports:** -0.023 (n.s.); **exit:** $+0.003$ (n.s.): no relocation, no differential closure.
- **TFPQ:** flat; no gain in physical productivity.

Unlike the advanced-economy benchmark, where regulation induced clean innovation (Aghion et al. 2016; Calel and Dechezleprêtre 2016), here the upgrading margin is too costly: plants comply by shrinking.

Conclusion

What margins do firms use to comply with environmental regulation? I explore this in the case of India's PAT scheme.

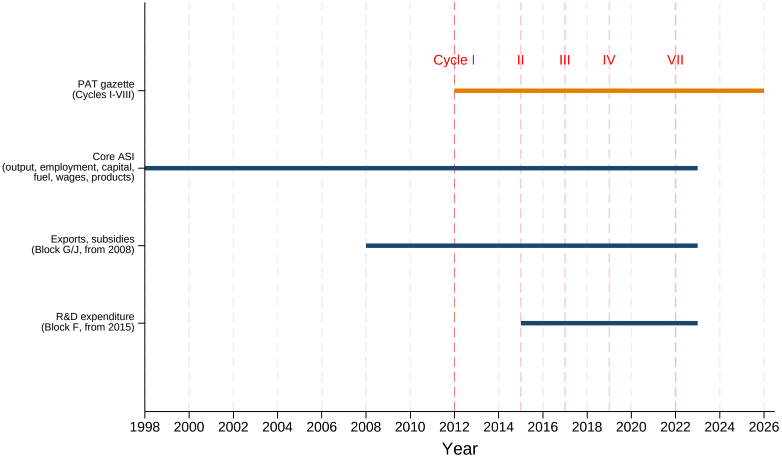
1. **Contraction, not innovation:** fuel falls 9–18%, capital and employment fall; R&D, investment, products, and unit values unchanged.
2. **Physical, not financial friction:** old plants contract most, financial access does not mitigate the contraction; energy and labor are complements.
3. **Misallocation:** the burden raises TFPR for fuel-intensive plants without raising TFPQ, adding revenue-productivity dispersion.

Contribution: first plant-level evaluation of a developing-country energy-intensity scheme; the compliance margin is contraction, not the clean innovation seen in advanced economies.

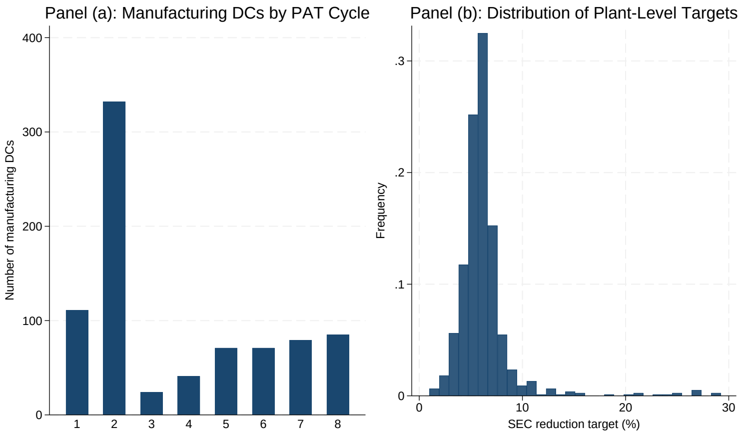
Policy (CCTS 2025): inherits PAT's intensity-based architecture; without affordable upgrading, expect further contraction.

Backup

Backup: Policy Timeline



Backup: PAT Cycles and Target Distribution



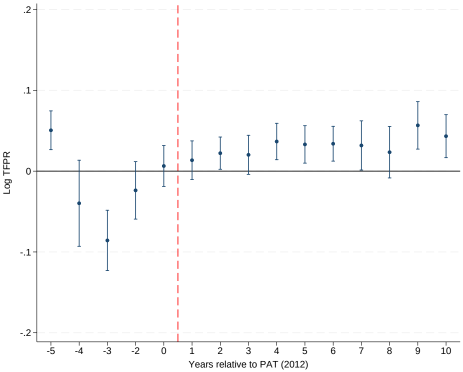
Panel (a): manufacturing Designated Consumers newly assigned per cycle (Cycle II the largest expansion, ∼330). Panel (b): distribution of plant-specific SEC reduction targets across all cycles. back to data

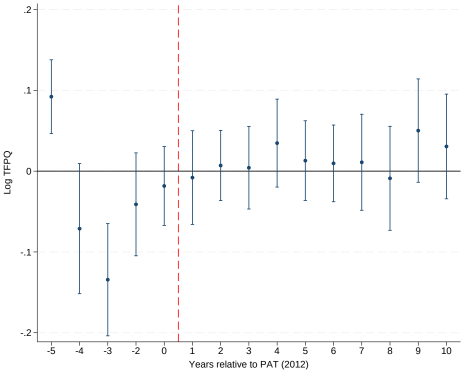
Backup: Dynamic Effects by Period

	(1) Log Fuels	(2) Log Fuel Int	(3) Log Capital	(4) Log Emp	(5) Log Output	(6) Log TFPQ
Early (2012–15)	−0.088***	−0.047***	−0.068***	−0.057**	−0.041**	−0.021***
	(0.019)	(0.013)	(0.023)	(0.025)	(0.020)	(0.006)
Mid (2016–19)	−0.137***	−0.131***	−0.078***	−0.089***	−0.006	−0.005
	(0.036)	(0.025)	(0.025)	(0.028)	(0.024)	(0.011)
Late (2020–23)	−0.177***	−0.197***	−0.076**	−0.122***	0.020	0.008
	(0.038)	(0.027)	(0.032)	(0.032)	(0.037)	(0.014)
Factory FE	Yes	Yes	Yes	Yes	Yes	Yes
Sector × Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	254,226	254,226	254,399	254,385	254,399	243,224

Notes: All dependent variables are in logs. Each cell reports the coefficient on PAT sector × standardized pre-period fuel intensity × period indicator; the pre-period (before 2012) is the omitted category. K = capital, Emp = employees, Y = output. TFPQ in column (6) is quantity TFP from the Hsieh-Klenow value-added framework (K, L, M); the pooled triple-difference using the Klenow-Pastén-Ruane gross-output measure gives TFPQ = −0.010 (p = 0.55), confirming no technology adoption across measurement choices. Sample includes all census-stratum factories (PAT and non-PAT). Factory FE absorb PAT sector status and baseline fuel intensity; sector×year FE absorb sector-level shocks. SE clustered at the factory level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

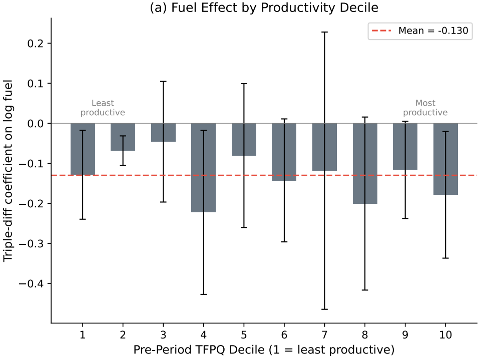
Backup: Event Study, log TFPR and log TFPQ

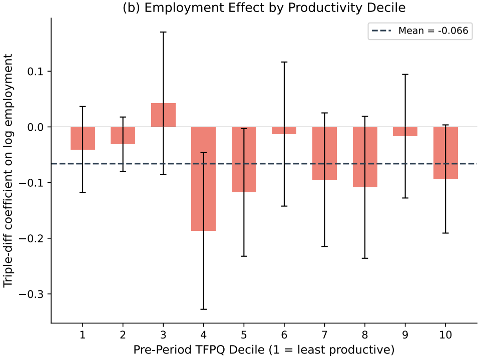




Notes: Triple-difference event-study coefficients on log TFPR and log TFPQ, both within-sector-year demeaned. Triple interaction: PAT-sector dummy $\times$ pre-PAT fuel intensity (standardized within sector) $\times$ event-time dummies. Endpoints binned at $k \leq -5$ and $k \geq +10$. Reference: 2011 ($k = -1$). Standard errors clustered at the plant level. Sample: ASI census stratum 1998–2023, $N = 254{,}201$.

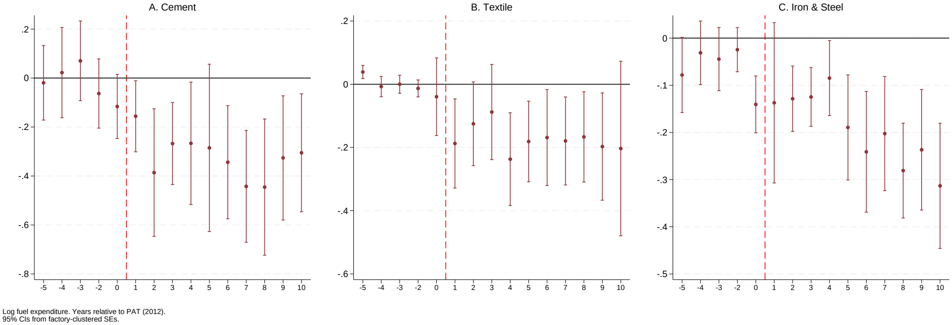
Backup: Treatment Effect by Productivity Decile





Notes: Each bar shows the triple-difference coefficient estimated separately for plants in each decile of pre-period TFPQ (within sector). Whiskers are 95% CIs. The dashed line marks the mean effect across all deciles. The flat pattern rules out the "zombie purging" interpretation: the regulation contracts productive and unproductive plants equally, conditional on energy intensity.

Backup: Event Study by Sector



Notes: Each panel plots sector-specific event-study coefficients for log fuel expenditure. The specification includes factory fixed effects and sector-by-year fixed effects. Base year is 2011 ($k = -1$). 95% CIs from factory-clustered standard errors.

Backup: Heterogeneous Effects by Proxies for c

	(1) Log Fuels	(2) Log Employment
Old plants (Q4 age) × PAT × $\mathrm{FI}_{\mathrm{pre}}$ × Post	−0.082	−0.132**
	(0.062)	(0.052)
High K/L (Q4) × PAT × $\mathrm{FI}_{\mathrm{pre}}$ × Post	+0.063	+0.121***
	(0.044)	(0.033)
Multi-plant firm × PAT × $\mathrm{FI}_{\mathrm{pre}}$ × Post	+0.049	+0.046
	(0.043)	(0.042)
Limited company × PAT × $\mathrm{FI}_{\mathrm{pre}}$ × Post	−0.020	−0.069
	(0.067)	(0.061)
Has R&D (2015+) × PAT × $\mathrm{FI}_{\mathrm{pre}}$ × Post	−0.065	+0.102
	(0.160)	(0.136)
Factory FE	Yes	Yes
Sector × Year FE	Yes	Yes

Notes: Each row reports the interaction of the proxy variable with the main triple-difference (PAT × $\mathrm{FI}_{\mathrm{pre}}$ × Post) from a separate regression. Positive coefficients mean plants with that characteristic contracted *less*. Proxies group into physical technology (Old plants, High K/L), financial access (Multi-plant firm, Limited company), and innovation capability (Has R&D). A separate test of the main effect on log gross investment yields +0.028 ($p = 0.23$). SE clustered at the factory level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Backup: Robustness and Additional Margins

	Log Fuels	Log Emp	Log Output
Panel A: COVID robustness			
Pre-COVID only (1998–2019)	−0.103***	−0.064**	−0.025
	(0.023)	(0.026)	(0.018)
Pre-COVID interaction (2012–19)	−0.105***	−0.068**	−0.030
	(0.024)	(0.027)	(0.019)
COVID interaction (2020–23)	−0.171***	−0.118***	0.016
	(0.037)	(0.032)	(0.036)
Panel B: Labor composition			
	Direct Emp	Contract Emp	Services Inc
PAT × FI_{pre} × Post	−0.081***	−0.052***	−0.112**
	(0.028)	(0.015)	(0.044)
Panel C: Extensive margin and technology adoption			
	Exit	R&D	Exports
PAT × FI_{pre} × Post	0.003	−0.019	−0.023
	(0.003)	(0.015)	(0.072)

Notes: All specifications include factory FE and sector×year FE. Standard errors clustered at the factory level. Panel A restricts to pre-COVID or adds period-specific interactions. Panel B decomposes employment into direct and contract workers and reports services income. Panel C tests the extensive margin (exit) and direct measures of technology adoption (R&D and exports); all three are null. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

References I

Philippe Aghion, Antoine Dechezleprêtre, David Hemous, Ralf Martin, and John Van Reenen. Carbon taxes, path dependency, and directed technical change: Evidence from the auto industry. *Journal of Political Economy*, 124(1):1–51, 2016.

Raphael Calel and Antoine Dechezleprêtre. Environmental policy and directed technological change: Evidence from the European carbon market. *Review of Economics and Statistics*, 98(1):173–191, 2016. URL `https://doi.org/10.1162/REST_a_00470`.

Andreas Fuster, Gazi Kabas, Vincenzo Pezone, and Kasper Roszbach. The labor market effects of carbon pricing. Working paper, EPFL, 2025.

Michael Greenstone, John A List, and Chad Syverson. The effects of environmental regulation on the competitiveness of U.S. manufacturing. *American Economic Review*, 102(6):2734–2768, 2012. URL `https://doi.org/10.1257/aer.102.6.2734`.

Rema Hanna. U.S. environmental regulation and FDI: Evidence from a panel of U.S.-based multinational firms. *American Economic Journal: Applied Economics*, 2(3):158–189, 2010.

Peter J Klenow, Ernesto Pastén, and Cian Ruane. Carbon taxes and misallocation in Chile. Working Paper, Stanford University, Central Bank of Chile, and Central Bank of Ireland, 2025.

References II

Ralf Martin, Mirabelle Muuls, Laure B de Preux, and Ulrich J Wagner. Industry compensation under relocation risk: A firm-level analysis of the EU emissions trading scheme. *American Economic Review*, 104(8):2482–2508, 2014.

W Reed Walker. The transitional costs of sectoral reallocation: Evidence from the Clean Air Act and the workforce. *Quarterly Journal of Economics*, 128(4):1787–1835, 2013.